

Lab Assignment 03, DDD: Bounded Contexts & Context Mapping

Application Scenario 4: Instant Mobility Platform

Team: The Winx | Summer Semester 2026

1 Overview

Domain-Driven Design (DDD) Strategic Design decomposes a complex system into Bounded Contexts, explicit boundaries within which a specific domain model applies and a consistent Ubiquitous Language is used. This document identifies the five bounded contexts of the Instant Mobility Platform and maps the relationships between them using Vernon's context mapping notation.

Each bounded context corresponds to one future microservice in the system architecture. Domain concepts that appear in multiple contexts are represented with different attributes in each context, only the data relevant to that context's responsibility is present.

2 Identified Bounded Contexts

Five bounded contexts were identified by grouping domain concepts around cohesive business capabilities with clear ownership boundaries and distinct ubiquitous languages.

ID	Context	Responsibility	Key Concepts
BC-01	Identity & Access	Authentication & account management for Users and Providers.	User (identity), Provider (identity)
BC-02	Fleet Management	CRUD, pricing, restrictions, and status tracking for vehicles.	Provider (owner), Vehicle, VehicleType, VehicleStatus, BillingModel
BC-03	Booking	Vehicle search, booking lifecycle, cost computation, payment trigger.	User (customer), Vehicle (availability), Booking, BookingStatus
BC-04	Payment	Financial transaction processing for completed bookings.	Booking (ref), Payment, PaymentStatus
BC-05	Rating	User feedback on vehicles and providers after completed bookings.	User (author), Vehicle (ref), Provider (ref), Booking (ref), Rating

2.1 Context Details

BC-01 Identity & Access (Purple)

Responsibility: Handles registration, authentication and credential management for both Users and Providers. Acts as the single source of truth for identity across the entire platform.

Domain Concept	Role in this context
User	Identity view, credentials, email, name, registeredAt. No booking or rating data lives here.
Provider	Identity view, credentials, companyName, contactName. No fleet data lives here.

Shared concepts: User and Provider appear in other contexts as well, but only their identity attributes (ID, email, name) are shared, not the full entity.

Ubiquitous Language:

- Registration, the act of creating a new account on the platform.
- Authentication, verifying that the supplied credentials match a stored account.
- Session, a time-limited proof that a principal has been authenticated.
- Principal, a generic term for any authenticated party (User or Provider).

BC-02 Fleet Management (Teal)

Responsibility: Manages the full lifecycle of vehicles offered by Providers: creation, updates, deletion, pricing configuration, restriction rules, and real-time availability and GPS location tracking.

Domain Concept	Role in this context
Provider	Fleet-owner view, which vehicles belong to this provider, fleet status overview.
Vehicle	Full entity, type, status, location, pricing, all restriction attributes.
VehicleType	Enum: E_SCOOTER BICYCLE E_BIKE E_CAR.
VehicleStatus	Enum: AVAILABLE BOOKED, updated on booking events from Booking context.
BillingModel	Enum: PER_HOUR PER_KILOMETER, set per vehicle by the Provider.

Shared concepts: Vehicle is shared into the Booking context (availability/location view) and into the Rating context (reference for what was rated).

Ubiquitous Language:

- Fleet, the collection of all vehicles owned by a single Provider.
- Availability, whether a vehicle can currently be booked.
- Restriction, a rule that constrains who may book a vehicle and for how long.
- Pricing, the combination of pricePerUnit and billingModel for a vehicle.

BC-03 Booking (Blue)

Responsibility: Owns the core rental lifecycle: searching for available vehicles by location, filtering by type and restrictions, creating a booking, tracking the active ride, computing the final cost on completion, and triggering payment.

Domain Concept	Role in this context
User	Customer view, who is making the booking. Only ID and age (for restriction checks) are used.
Vehicle	Availability view, current location, status, pricePerUnit, billingModel, restrictions. Read from Fleet Management.
Booking	Full entity, start/end times, GPS coordinates, distanceKm, totalCost, status.
BookingStatus	Enum: ACTIVE COMPLETED CANCELLED.

Shared concepts: Booking is shared (read-only reference) into the Payment context (to trigger and settle payment) and into the Rating context (to check that a booking is COMPLETED before a rating is allowed).

Ubiquitous Language:

- Search, finding vehicles within a given radius of the user's position.
- Booking, a confirmed reservation of a vehicle for immediate use.
- Active ride, the period between booking start and booking end.
- Cost computation, deriving totalCost from pricePerUnit, billingModel, duration, and distance.

BC-04 Payment (Amber)

Responsibility: Processes and records financial transactions for completed bookings. Receives a payment trigger from the Booking context, charges the user, and records the outcome (PAID or FAILED). Owns no booking logic.

Domain Concept	Role in this context
Booking	Reference only, bookingId and totalCost are used to create the Payment record.
Payment	Full entity, amount, paymentMethod, paidAt, status.
PaymentStatus	Enum: PENDING PAID FAILED.

Shared concepts: No domain concepts from Payment are shared outward. Payment is purely a downstream consumer of Booking.

Ubiquitous Language:

- Payment, a financial transaction tied to one completed booking.
- Charge, the act of debiting the user's chosen payment method.
- Settlement, confirmation that the payment has been successfully processed.

BC-05 Rating (Coral / Red)

Responsibility: Collects and stores user feedback (scores and comments) for a Vehicle and its Provider after a booking reaches COMPLETED status. Enforces the rule that only one rating is allowed per booking.

Domain Concept	Role in this context
User	Author view, who wrote the rating. Only userId and name are used.
Vehicle	Rated subject, vehicleId and description used for display. Read from Fleet Management.
Provider	Rated subject, providerId and companyName. Read from Fleet Management.
Booking	Reference, bookingId used to validate that the booking is COMPLETED and that no rating exists yet.
Rating	Full entity, vehicleScore, providerScore, comment, createdAt.

Shared concepts: No domain concepts from Rating are shared outward.

Ubiquitous Language:

- Rating, a user's evaluation of a vehicle and provider after a completed ride.
- Score, a numeric value (1–5) representing quality.
- Review, the optional free-text comment accompanying the scores.

3 Context Map (Vernon's Notation)

The context map below documents all integration relationships between the five bounded contexts. For each relationship the upstream (U) and downstream (D) context are identified, and one or more context mapping patterns from Vaughn Vernon's DDD are applied.

3.1 Pattern Reference

Abbreviation	Pattern	Meaning
OHS	Open Host Service	The upstream context publishes a well-defined, stable service protocol (API) that any downstream can consume. Avoids tight coupling to internal implementation details.
PL	Published Language	A well-documented shared exchange format (e.g. JSON schema, protobuf) used by the OHS. Allows downstream consumers to integrate without knowing the upstream's internal model.
ACL	Anti-Corruption Layer	A translation layer in the downstream context that converts the upstream model into the downstream's own domain model. Protects the downstream from upstream changes and keeps the domain clean.

c/s	Customer / Supplier	A negotiated relationship where the downstream (customer) can influence the upstream (supplier) interface through planning discussions. The supplier gives priority to the customer's needs.
CF	Conformist	The downstream adopts the upstream model without translation. Chosen when the upstream model is stable and well-designed enough that translation would add cost without benefit.

3.2 Relationships

Upstream (U)	Downstream (D)	U-Pattern	D-Pattern	Rationale
Identity & Access	Fleet Management	OHS / PL	ACL	Fleet Management needs to verify Provider identity before allowing vehicle management operations. Identity & Access exposes a stable Open Host Service with a Published Language (e.g. JWT token or REST endpoint). Fleet Management uses an Anti-Corruption Layer to translate the IAC identity model into its own Provider concept.
Identity & Access	Booking	OHS / PL	ACL	Booking must verify User identity and retrieve age for restriction checks. Identity & Access provides the same OHS/PL. Booking uses an ACL to translate the IAC User identity into its own lightweight customer representation.
Identity & Access	Rating	OHS / PL	ACL	Rating needs the authenticated User's identity (userId, name) to associate a rating with its author. Identity & Access provides OHS/PL; Rating uses an ACL to map the identity data to its own author concept.
Fleet Management	Booking	OHS / PL	ACL	Booking queries Fleet Management for available vehicles (by location, type, and restrictions) and reads vehicle pricing data to compute ride costs. Fleet Management publishes a stable OHS with a clearly defined API. Booking uses an ACL so that internal changes to Fleet Management's vehicle model do not propagate into Booking's domain model.
Booking	Payment	Customer / Supplier (C/S)	Conformist (CF)	After a Booking is COMPLETED, the Booking context publishes an event (or makes a call) to trigger payment. This is a negotiated Customer/Supplier relationship, Booking (supplier) and Payment (customer) collaborate on the interface. Payment is Conformist: it accepts and uses Booking's data

				structure (bookingId, totalCost) without translation, simplifying integration.
Booking	Rating	Customer / Supplier (C/S)	Conformist (CF)	Rating can only be submitted for a COMPLETED booking and must verify there is no existing rating for that booking. Booking is the supplier; Rating is the customer. Rating is Conformist, it uses Booking's BookingStatus enum and bookingId directly rather than maintaining its own copy of booking state.
Fleet Management	Rating	OHS / PL	ACL	Rating needs stable references to the Vehicle and Provider that are being rated (for display names and IDs). Fleet Management exposes these via its OHS. Rating uses an ACL to translate them into lightweight rated-subject objects, decoupling Rating from future changes to Fleet Management's vehicle or provider model.

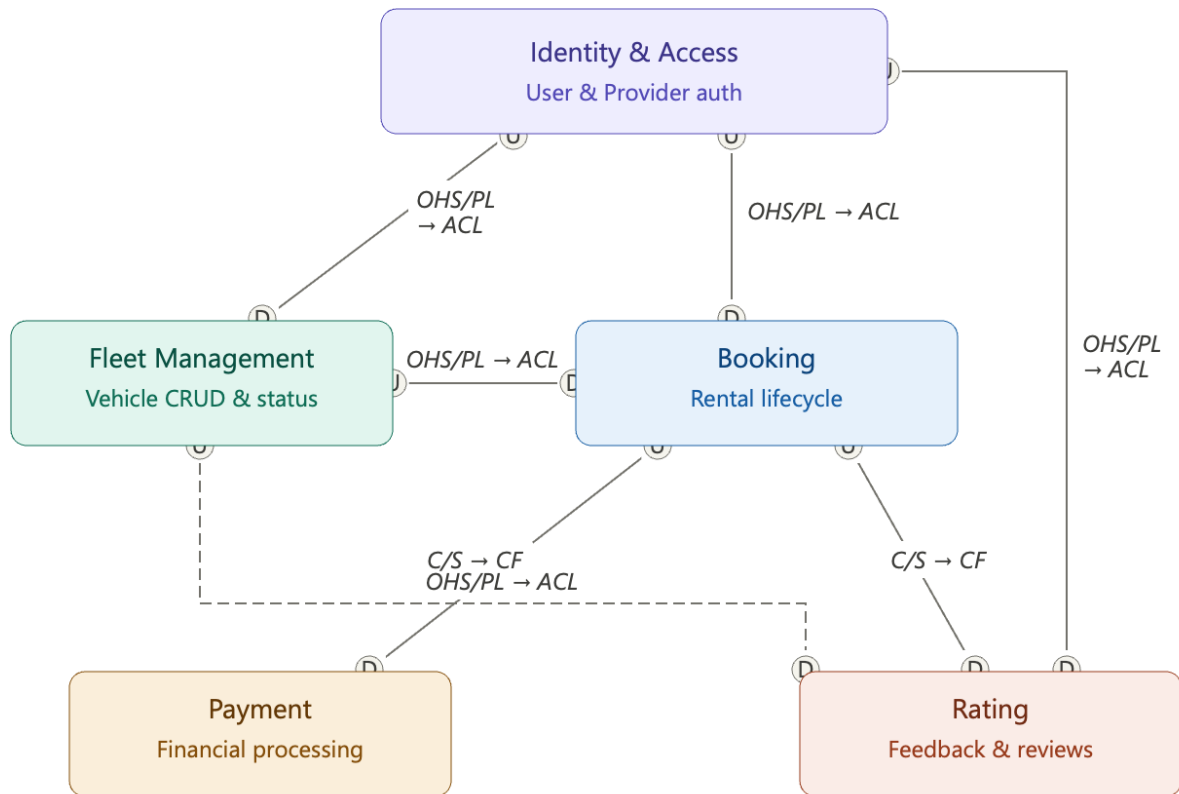
3.3 Context Map Narrative

Identity & Access is the global upstream for the entire platform. Every other context authenticates against it using the OHS/PL it provides, wrapping the result in an ACL so that identity changes never pollute domain models. This single source of truth prevents each context from re-implementing authentication logic.

Fleet Management sits in an upstream position relative to Booking and Rating. It publishes vehicle and provider data via an OHS. Booking uses this to locate available vehicles and to retrieve pricing for cost computation. Rating uses it to obtain display names for the rated vehicle and provider. Both protect themselves via an ACL.

Booking is the central operational context, it orchestrates the ride lifecycle. It is simultaneously a downstream consumer (of IAC and Fleet Management) and an upstream supplier (to Payment and Rating). The relationship with Payment and Rating is Customer/Supplier: both downstream contexts had input into the interface during design. Payment and Rating are Conformist because Booking's output model (bookingId, totalCost, BookingStatus) is stable and well-defined.

Payment and Rating are pure leaf contexts, they consume from others and produce no data that is consumed by any other context in the system.



Context mapping decisions:

$OHS/PL \rightarrow ACL$ used where the upstream is a stable platform service (Identity & Access, Fleet Management). The ACL on the downstream side means internal upstream changes never leak into other contexts.

Customer/Supplier $\rightarrow$ Conformist used between Booking $\rightarrow$ Payment and Booking $\rightarrow$ Rating because both downstream contexts had a say in the interface design (C/S), and Booking's output data is clean enough that no translation layer is needed (CF).